Nijat Mahmudov

Electrical Engineering B.Sc. (in progress)
Istanbul, Türkiye | (+90) 507 972 8177 | mahmudov21@itu.edu.tr

PROFESSIONAL SUMMARY

Electrical Engineering undergraduate at Istanbul Technical University with experience in power quality systems and industrial automation environments. Skilled in modeling and design, including DC/DC converter analysis and implementation workflows. Experience with SCADA exposure, PLC-driven systems, panel design, and power system compensation practices for industrial applications. Strong interest in embedded systems design for real-world cyber-physical infrastructures.

EDUCATION

Istanbul Technical University (ITU) — Istanbul, Türkiye
B.Sc. Electrical Engineering — *Sep 2021 – Present*

- **Current GPA:** 3.21/4.00
- **Relevant Interests:** Power electronics, industrial control, smart grids, real-time systems.

ACADEMIC PROJECTS

Graduation Project: Equivalent Modeling of High Penetration Distribution Networks

- Studied the impact of high renewable energy penetration on distribution grids.
- Developed equivalent circuit models to analyze grid stability and power flow.

WORK EXPERIENCE

ARMES Group — Istanbul, Türkiye
Electrical Engineer (Power Quality & Industrial Systems) — *July 2025 – Feb 2, 2026*

- Developed and tested power quality solutions including SVG/ASVG, active harmonic filters, and UPS systems for industrial clients.
- Performed harmonic analysis and power factor correction planning.
- Modeled and analyzed DC/DC converter stages in MATLAB/Simulink and LTspice.
- Worked with industrial monitoring/control contexts: SCADA exposure, PLC-based field integrations, and instrumentation interfaces in real installations.
- Supported LV/MV electrical panel workflows including panel design considerations, wiring review, and commissioning documentation.

SELECTED TECHNICAL FOCUS

- **Power Electronics:** DC/DC converter modeling, control-relevant behavior, and validation workflows.
- **Industrial Systems:** SCADA/PLC contexts, field devices, and panel-level integration.
- **Power Quality:** Harmonic mitigation, reactive power support, compensation planning, and industrial client deployments.

SKILLS

Engineering:	Power Electronics, Power Quality (SVG/ASVG, Active Harmonic Filters), Industrial Panels, Compensation.
Control:	Real-time system mindset, converter-level modeling, signal integrity awareness, industrial integration.
Software/Tools:	MATLAB/Simulink, C, LTspice, ETAP, Microsoft Office.

LANGUAGES

- **Azerbaijani:** Native
- **English:** Near-native

- **Russian:** Near-native
- **Turkish:** Advanced / Upper level

VOLUNTEERING

Formula 1 Baku Baku, Azerbaijan
Volunteer *Jul 2023*